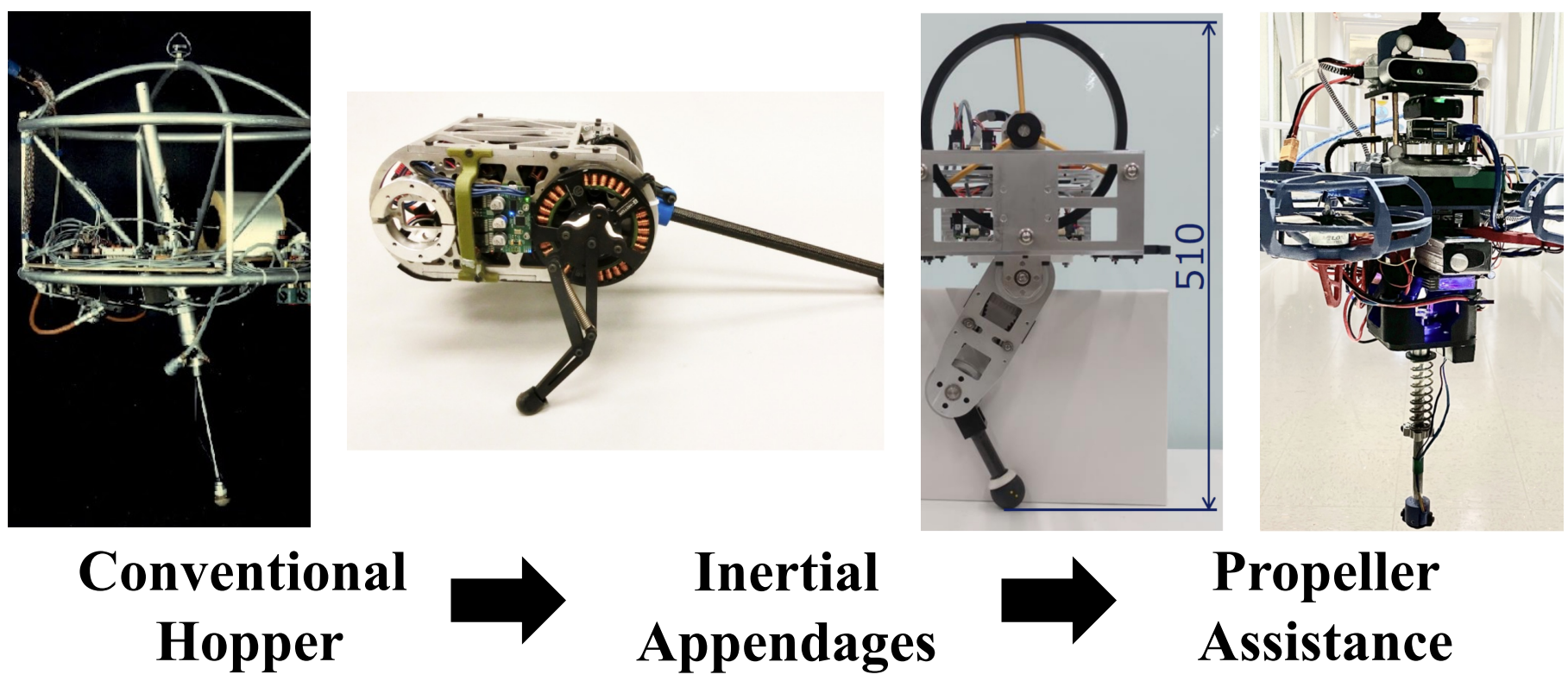


Propeller-Assisted Robust 3D Hopping Robot with Hierarchical Force Allocation

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1 Problem- hybrid dynamics create an authority gap

**Conventional Hopper** → **Inertial Appendages** → **Propeller Assistance**

■ Stance

Ground reaction force exists only during a short contact interval.

■ Flight

No contact: attitude becomes underactuated.

■ Disturbance

Impacts and pushes inject attitude and horizontal momentum.

Control authority disappears exactly when attitude can keep drifting.

Design question: how should leg contact force and rotor thrust be coordinated across stance and flight?

2 First proof- stable on 30° grass

Propellers OFF



Propellers ON



■ Propellers off

After asymmetric contact, attitude error grows. With no flight-phase actuator, the robot falls.

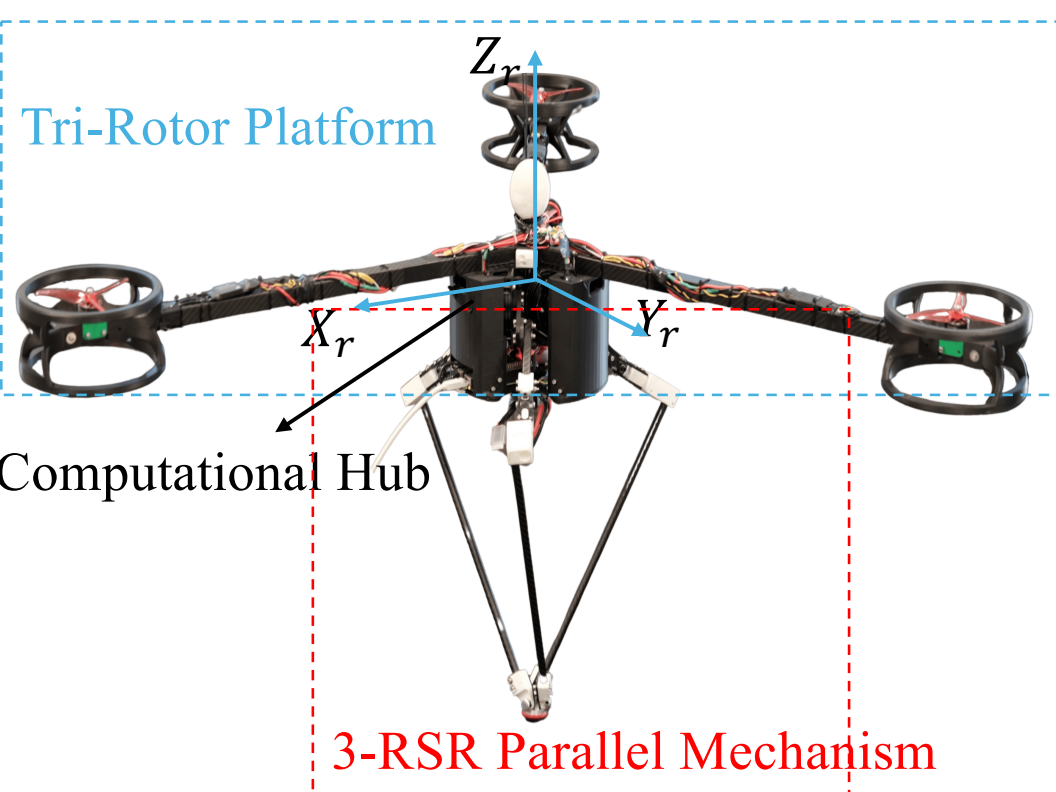
■ Propellers on

Differential thrust remains available through flight and restores attitude before the next landing.

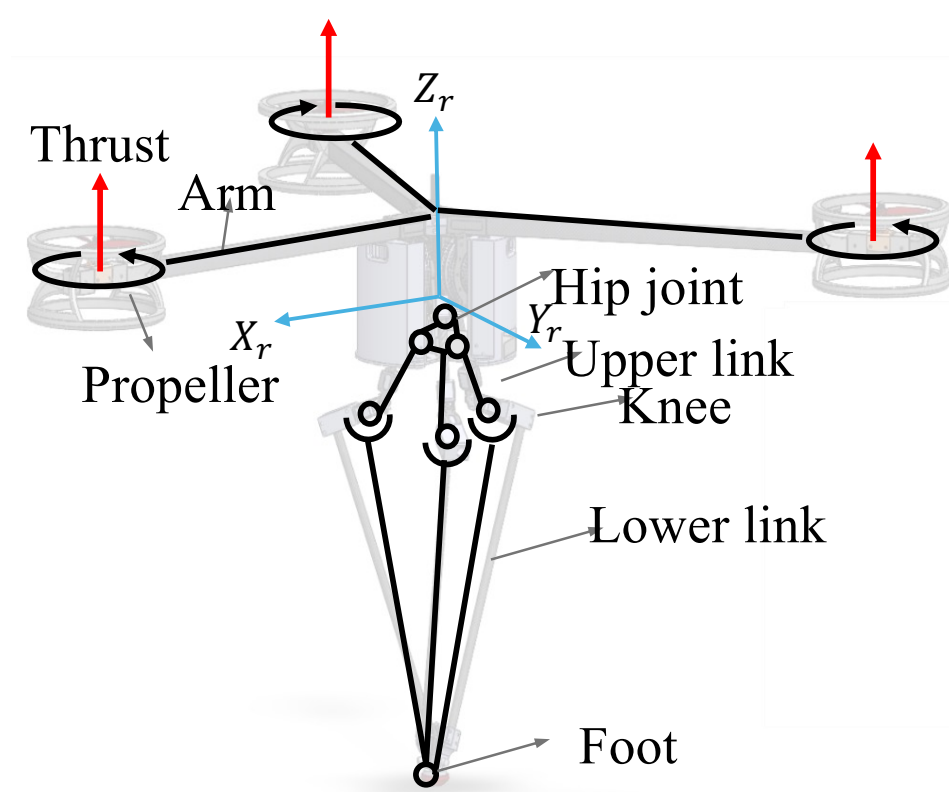
■ What this establishes

Residual authority in stance.**Primary** attitude authority in flight.**Faster recovery** from uneven impacts.**Not primary lift:** the rotors close the attitude-authority gap.**Propellers off:** attitude diverges and the robot falls. **Propellers on:** it remains upright and keeps hopping.

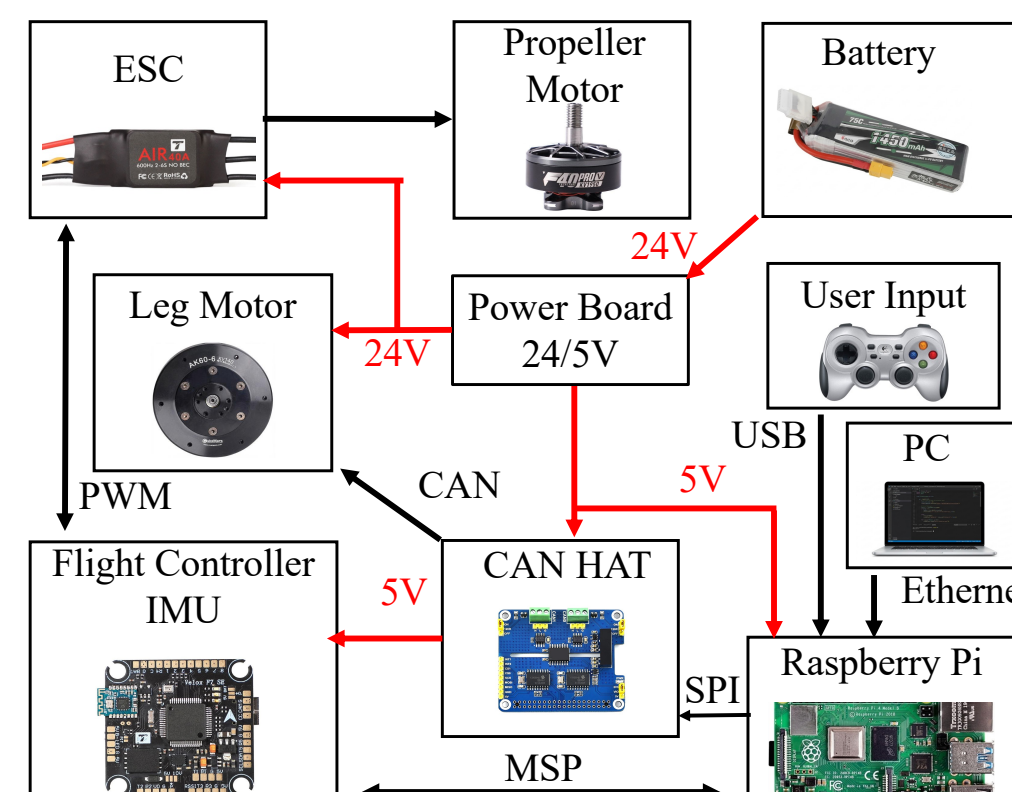
3 Platform- hardware built for leg-first hopping



Robot design



Leg and tri-rotor components



Electronic architecture

■ 3-RSR parallel leg

Three RSR chains meet at one foot: fast 3-DoF omnidirectional placement, only **0.35 kg** moving mass.

■ Computational hub

Batteries and electronics near the hip: low center of mass, low inertia.

■ Y-layout tri-rotor

Continuous attitude authority; **10 N** per rotor, total 0.82 mg: a hopper, not a drone.

■ External PC @ 500 Hz

State machine, estimation, wrench planning, and the HFA allocator.

■ Raspberry Pi interface

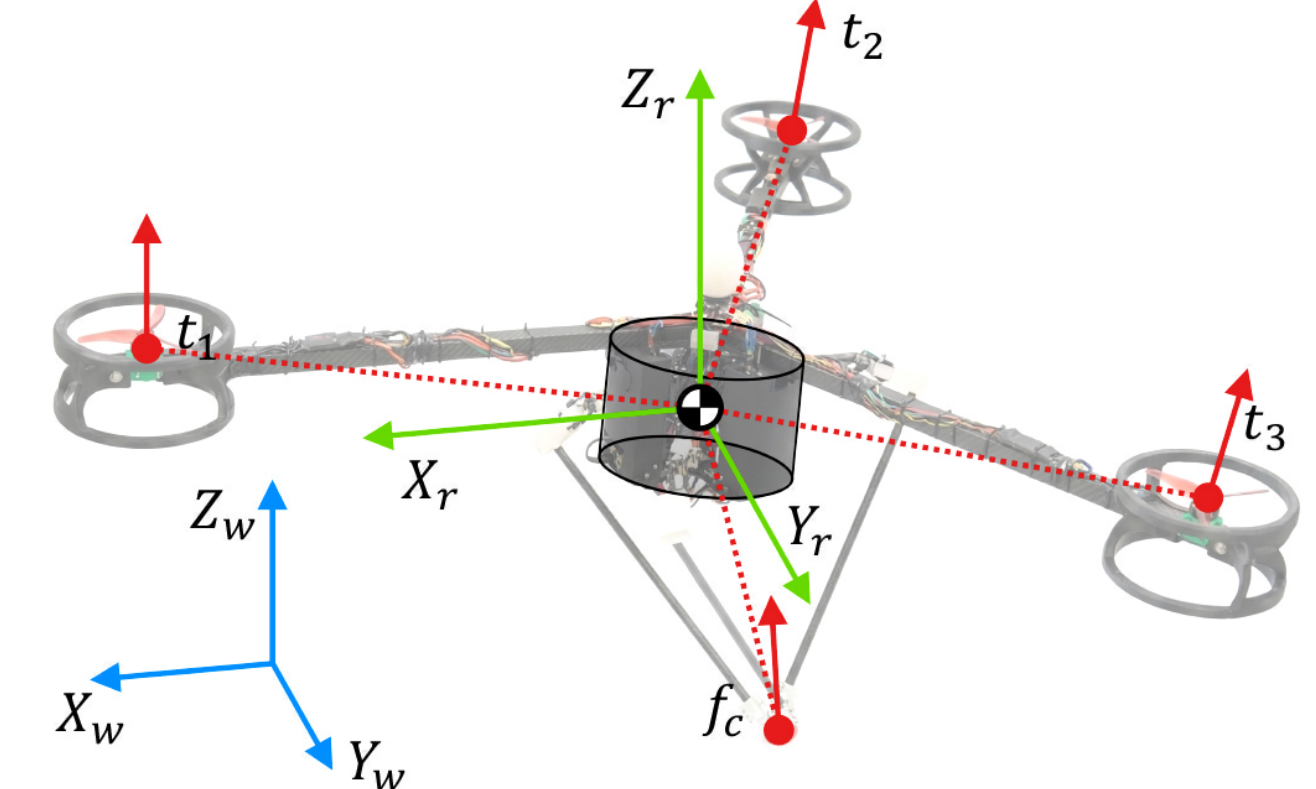
CAN to the three leg motors; MSP to the flight controller.

■ Real-time link

LCM over Ethernet: RTT ≈ 0.2 ms, 99% < 0.5 ms, within the 2 ms period.

3.75 kg total ■ 9.2% leg mass ■ 20 Nm / leg motor ■ 500 Hz control

4 Model- one rigid body, two actuator sets

Contact force f_c and thrusts $t_{1,2,3}$ produce the body wrench.

■ Net wrench = leg contact + three rotors

$$\begin{bmatrix} F_{net} \\ \tau_{net} \end{bmatrix} = \begin{bmatrix} f_c \\ r_c \times f_c \end{bmatrix} + \sum_{i=1}^3 \begin{bmatrix} d_i \\ r_i \times d_i \end{bmatrix} t_i$$

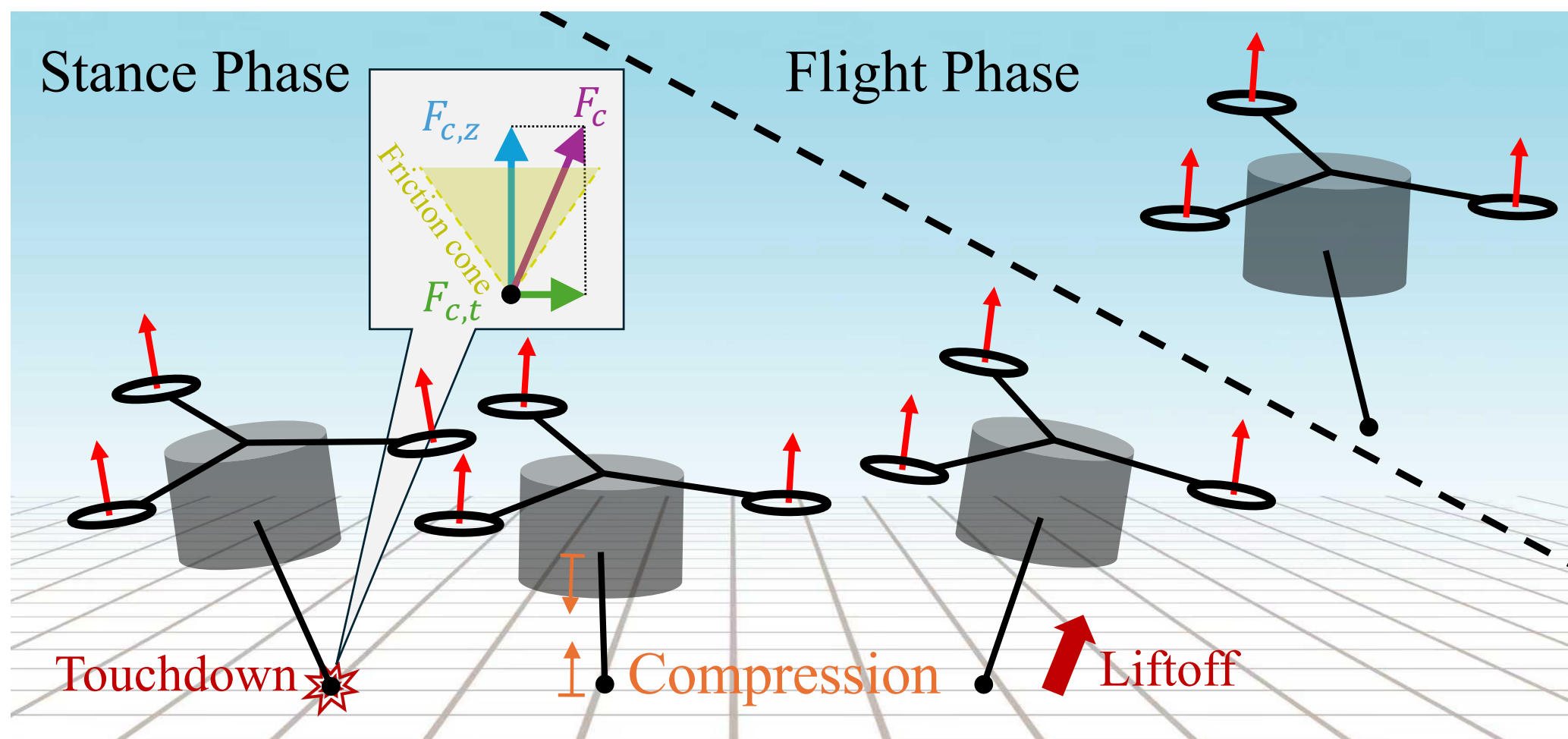
■ Newton-Euler dynamics of the CoM

$$ma = F_{net} + mg, \quad \dot{\omega} = I^{-1}(R^T \tau_{net} - \omega \times I \omega)$$

■ Symbols (values used in all experiments)

 f_c : ground reaction force (world) $m=3.75$ kg; g gravity; I inertia r_c, r_i : arms: CoM → contact, rotor i $R \in SO(3)$ attitude; ω body rate $t_i \in [0, 10]$ N thrust along $d_i = Re_3$ J leg Jacobian; $\tau_{max}=20$ Nm limit F_{net}, τ_{net} net force, moment $\mu=0.4$ friction coefficient $p_z, \dot{p}_z$ CoM height, velocity; $h_{des}=0.664$ m l leg length; $l_0=0.464$ m; Δ_{td} threshold k_z, b_z, k_E vertical, k_v, k_r Raibert gains K_R, K_ω attitude PD gains

5 Onehop- event-driven authority handoff



■ Touchdown: $l \leq l_0 - \Delta_{td}$

Leg absorbs impact; energy law drives the next apex.

■ Liftoff: $l \geq l_0$

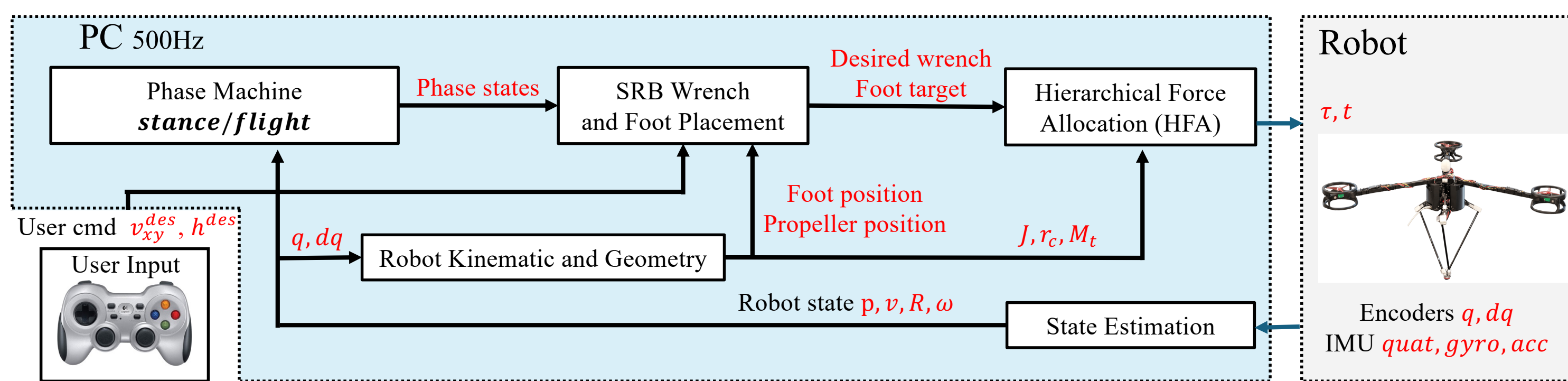
Contact wrench vanishes; rotors primary for attitude.

■ Handoff

Events switch authority; no hardware or gain change.

Leg authority exists only in stance; rotor attitude authority is continuous through the full hop.

6 Control- leg-priority hierarchical force allocation



500 Hz architecture: commands + feedback → planner → two-stage allocator → leg torques + rotor thrusts.

■ A. Vertical energy hopping (stance)

$$E_{sys} = \frac{1}{2} m \dot{p}_z^2 + mg p_z, \quad E_{des} = mg h_{des}$$

$$f_{c,z}^{des} = \max(0, k_z(h_{des} - p_z) - b_z \dot{p}_z + k_E(E_{des} - E_{sys}))$$

Injects push-off energy, absorbs impact.

■ C. Leg first- constrained least squares

$$f_{c,xy}^{cmd} = \arg \min_{f_{xy}} \| (r_c \times f_c)_{rp} - (\tau_{res}^{des})_{rp} \|_2^2$$

$$\text{s.t. } \|f_{xy}\|_2 \leq \mu f_{c,z}^{des}, \quad \|J^T(-R^T f_c)\|_\infty \leq \tau_{max}$$

The leg realizes all it can under friction and torque limits.

■ B. SO(3) attitude PD (both phases)

$$e_R = \frac{1}{2} (R_{des}^T R - R^T R_{des})^\vee$$

$$\tau_b^{des} = -K_R e_R + K_\omega (\omega^{des} - \omega)$$

Roll/pitch only; yaw left free. $\tau^{des} = R \tau_b^{des}$.

■ D. Rotors take only the residual

$$\tau_{res}^{cmd} = \tau^{des} - r_c \times f_c^{cmd}$$

$$\delta t^{cmd} = \arg \min_{\delta t} \| (M_t \delta t)_{rp} - (\tau_{res}^{cmd})_{rp} \|_2^2$$

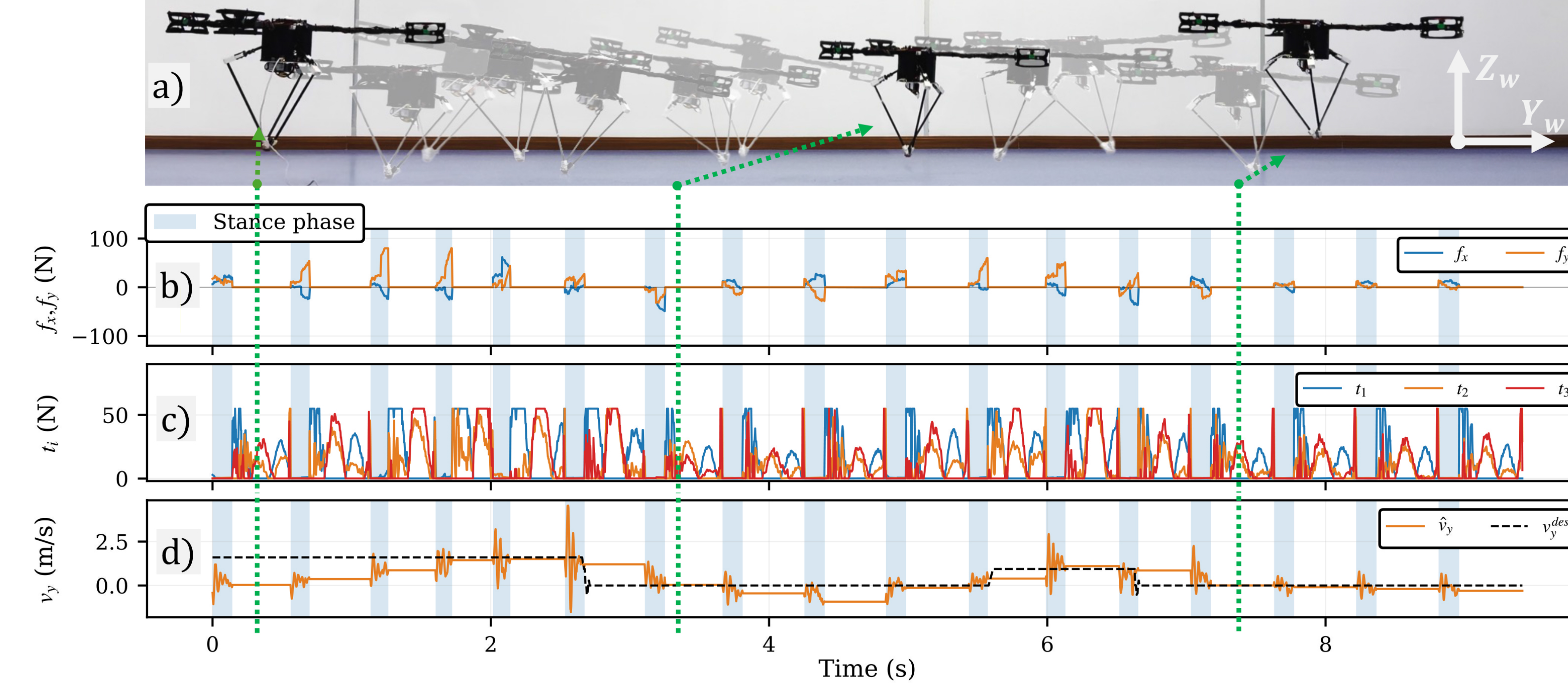
s.t. thrust bounds; $t^{cmd} = \frac{T_{base}}{3} \mathbf{1}_3 + \delta t^{cmd}$, $T_{base} = 0.02$ mg.

■ E. Flight- rotors primary; Raibert-style foot placement

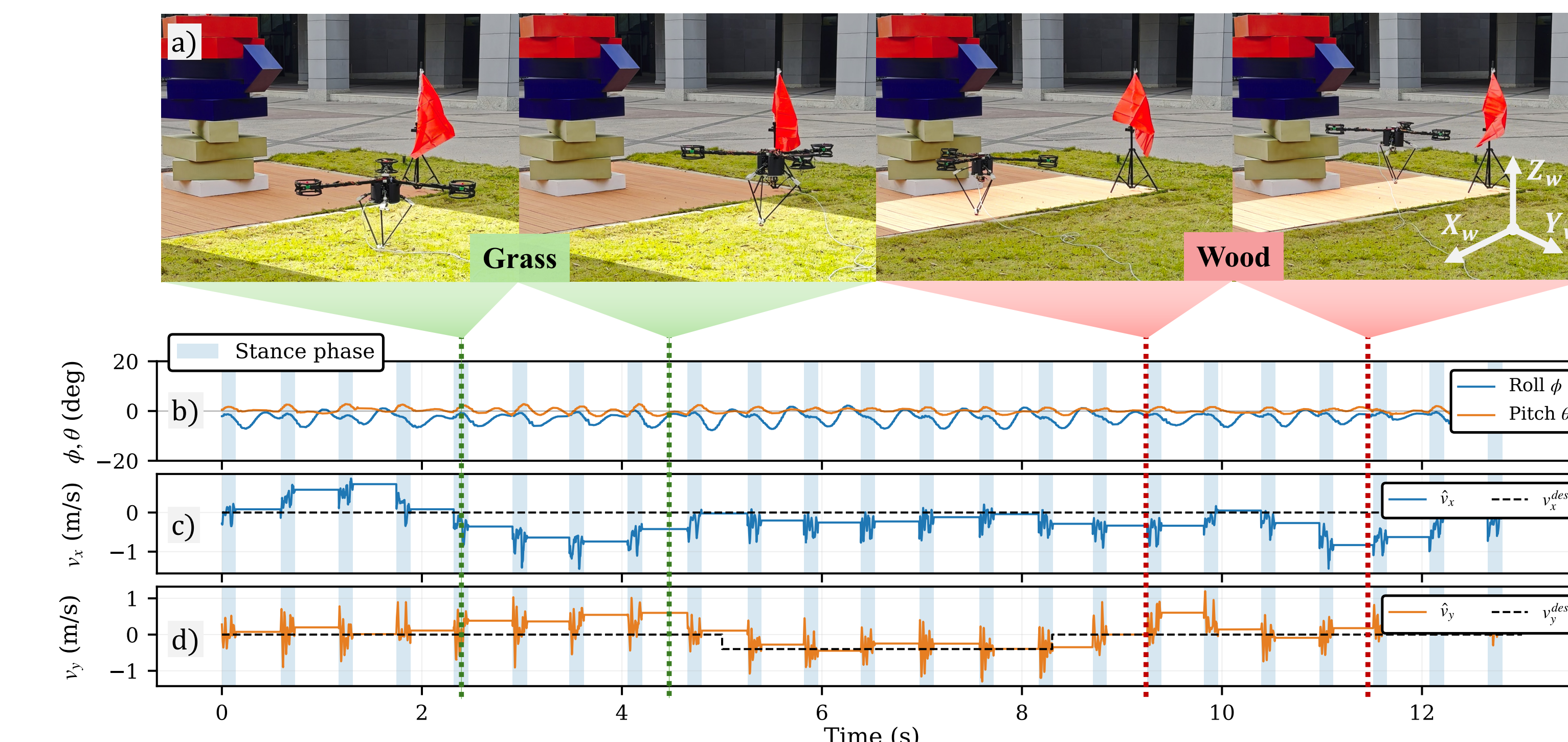
$$f_c = 0; \quad \text{rotors track } (\tau_{res}^{des})_{rp}; \quad r_{td,xy}^{des} = k_v v_{xy} - k_r v_{xy}^{des}$$

$$p_{foot}^{des} = [x_{td}^{des}, y_{td}^{des}, -\sqrt{l_0^2 - \|r_{td,xy}^{des}\|_2^2}]^T$$

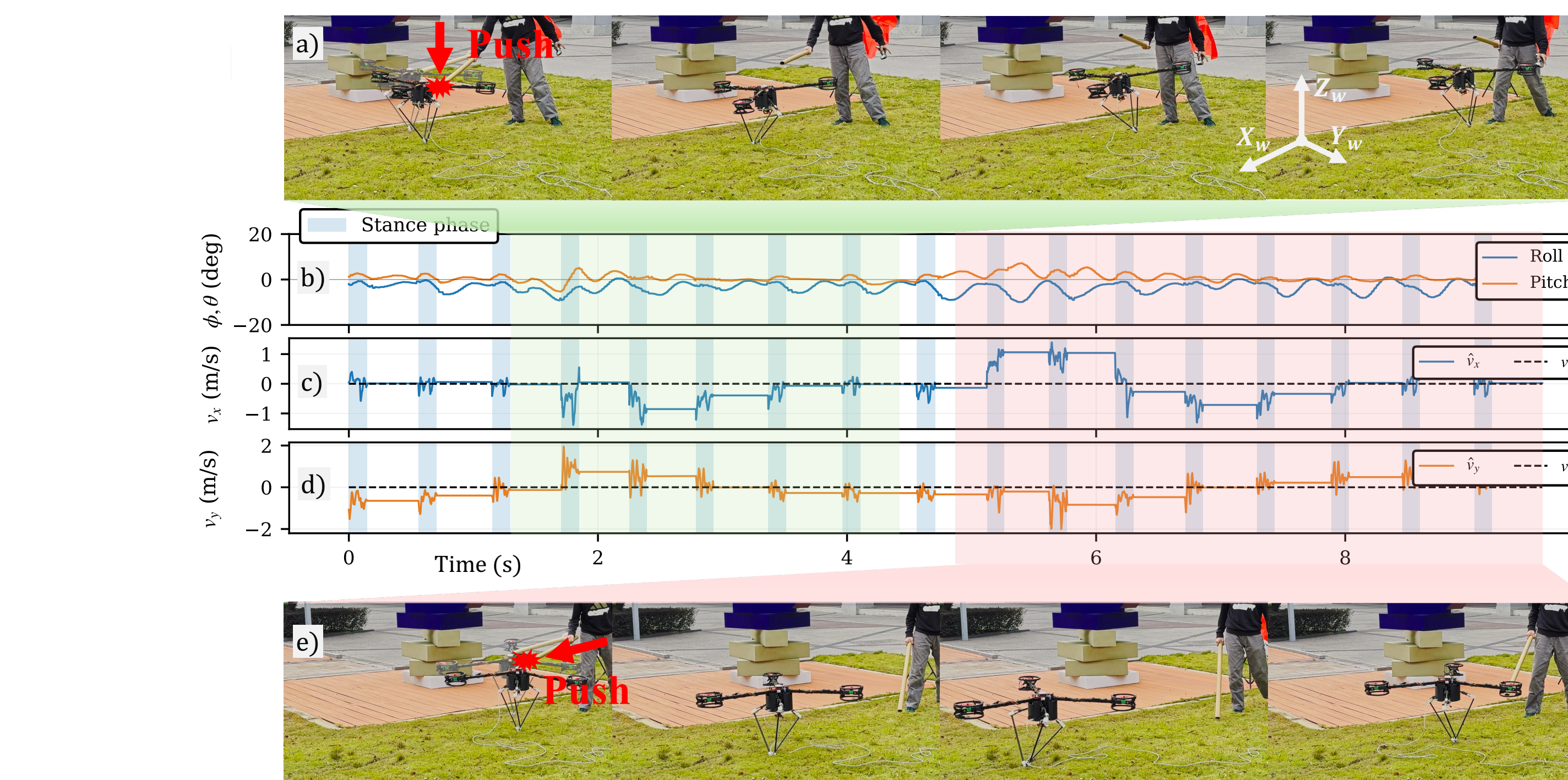
7 Validation- indoor stop-and-go tracking

Stop-and-go commands: accelerate, then hard stop (peaks 1.6 and 0.94 m/s) - 17 hops. Velocity recovered stride to stride by **Raibert-style foot placement** (Panel 6, E).

8 Validation- grass to wood

Grass → wood under light wind: **23 hops**- same controller, no terrain model, no gain retuning. Attitude stays tight: roll $\sigma = 2.1^\circ$, pitch $\sigma = 1.0^\circ$; deviations decay within 1-2 hops.

9 Validation- push recovery

Rotor-arm push: propellers restore roll/pitch in flight - stable again in **1-2 hops**. Body push (≈1.4 m/s velocity spike): foot placement recovers tracking in **3 hops**.